

Linear mixed effects models

Warmup

Work on the warmup activity (handout).

Warmup

What are the groups in the data?

Musicians

id	<u>na</u>	<u>large</u>
1	3	1
1	4	0
1	1	1

Warmup

What mixed effects model should the researchers use to investigate their hypothesis?

$$\underbrace{\text{Anxiety}_{ij}}_{\text{anxiety for musician } i \text{ before performance } j} = \beta_0 + \beta_1 \text{large}_{ij} + \underbrace{u_i}_{\substack{\uparrow \\ \text{variability} \\ \text{in anxiety} \\ \text{between} \\ \text{musicians}}} + \underbrace{\varepsilon_{ij}}_{\substack{\uparrow \\ \text{variability} \\ \text{in anxiety} \\ \text{for the} \\ \text{same musician,} \\ \text{before different} \\ \text{performances}}}$$
$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2)$$
$$\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

+ Shape:

- + the overall relationship between satisfaction and price is linear
- + The slope is the *same* for each neighborhood

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

+ Independence:

- + random effects are independent
- + observations within neighborhoods are independent after accounting for the random effect (i.e., the random effect captures the correlation within neighborhoods)
- + observations from different neighborhoods are independent

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- + **Normality:** Both $u_i \sim N(0, \sigma_u^2)$ and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$
- + **Constant variance:**
 - + ε_{ij} has the same variance σ_ε^2 regardless of satisfaction or neighborhood

Calculating predicted values

Residuals: $Price_{ij} - \widehat{Price}_{ij}$, where

$$\widehat{Price}_{ij} = \hat{\beta}_0 + \hat{\beta}_1 Satisfaction_{ij} + \hat{u}_i$$

Where do the $\hat{u}_i$ come from?

Estimated random effects

R calculates an estimated random effect for each group (i.e., neighborhood):

```
m1 <- lmer(price ~ overall_satisfaction +  
            (1 | neighborhood),  
            data = bnb)  
coef(m1)
```

```
## $neighborhood  
##               (Intercept) overall_satisfaction  
## Albany Park           16.367331           14.80912  
## Archer Heights         9.863461           14.80912  
## Avondale              40.533851           14.80912  
## Beverly               21.046464           14.80912  
## Bridgeport            13.304517           14.80912  
## Brighton Park         28.548742           14.80912  
## Burnside              12.741349           14.80912  
## Calumet Heights       11.465091           14.80912
```

Estimated random effects

```
coef(m1)
```

	(Intercept)	overall_satisfaction
Albany Park	16.367331	14.80912
Archer Heights	9.863461	14.80912
Avondale	40.533851	14.80912
Beverly	21.046464	14.80912
Bridgeport	13.304517	14.80912

What is the same for every neighborhood?

Estimated random effects

```
coef(m1)
```

	(Intercept)	overall_satisfaction
Albany Park	16.367331	14.80912
Archer Heights	9.863461	14.80912
Avondale	40.533851	14.80912
Beverly	21.046464	14.80912
Bridgeport	13.304517	14.80912

What is the same for every neighborhood?

The slope ($\hat{\beta}_1$)

Estimated random effects

```
coef(m1)
```

	(Intercept)	overall_satisfaction
Albany Park	16.367331	14.80912
Archer Heights	9.863461	14.80912
Avondale	40.533851	14.80912
Beverly	21.046464	14.80912
Bridgeport	13.304517	14.80912

What is *different* for each neighborhood?

Estimated random effects

$$\beta_0 + \beta_1 \text{Satisfaction}_{ij} + u_i + \varepsilon_{ij}$$

$(\beta_0 + u_i)$ + $\beta_1 \text{Satisfaction}_{ij}$ + ε_{ij}
intercept for neighborhood i

```
coef(m1)
```

	(Intercept)	overall_satisfaction
Albany Park	16.367331	14.80912
Archer Heights	9.863461	14.80912
Avondale	40.533851	14.80912
Beverly	21.046464	14.80912
Bridgeport	13.304517	14.80912

What is *different* for each neighborhood?

The intercept $(\hat{\beta}_0 + \hat{u}_i)$

Estimated random effects

	(Intercept)	overall_satisfaction
Albany Park	16.367331	14.80912
Archer Heights	9.863461	14.80912
Avondale	40.533851	14.80912
Beverly	21.046464	14.80912
Bridgeport	13.304517	14.80912

How do I get the random effect estimates $\hat{u}_i$?

Estimated random effects

	(Intercept)	overall_satisfaction
Albany Park	16.367331	14.80912
Archer Heights	9.863461	14.80912
Avondale	40.533851	14.80912
Beverly	21.046464	14.80912
Bridgeport	13.304517	14.80912

How do I get the random effect estimates $\hat{u}_i$?

The intercept for each neighborhood is $\hat{\beta}_0 + \hat{u}_i$

So subtract the estimate $\hat{\beta}_0$ from the intercept for each neighborhood

Checking assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

+ Normality assumption: Both $u_i \sim N(0, \sigma_u^2)$ and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

How do you think we could check the normality assumption?

QQ plots

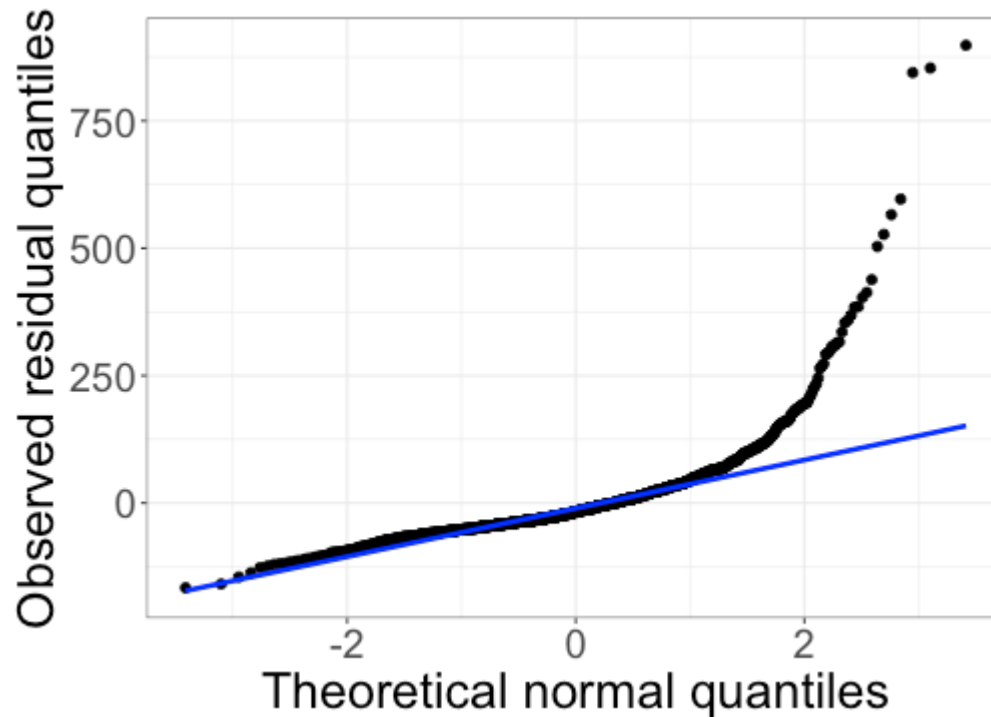
Assumption: $u_i \sim N(0, \sigma_u^2)$

- + Check whether the random effect estimates $\hat{u}_i$ appear normal with a QQ plot

Assumption: $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

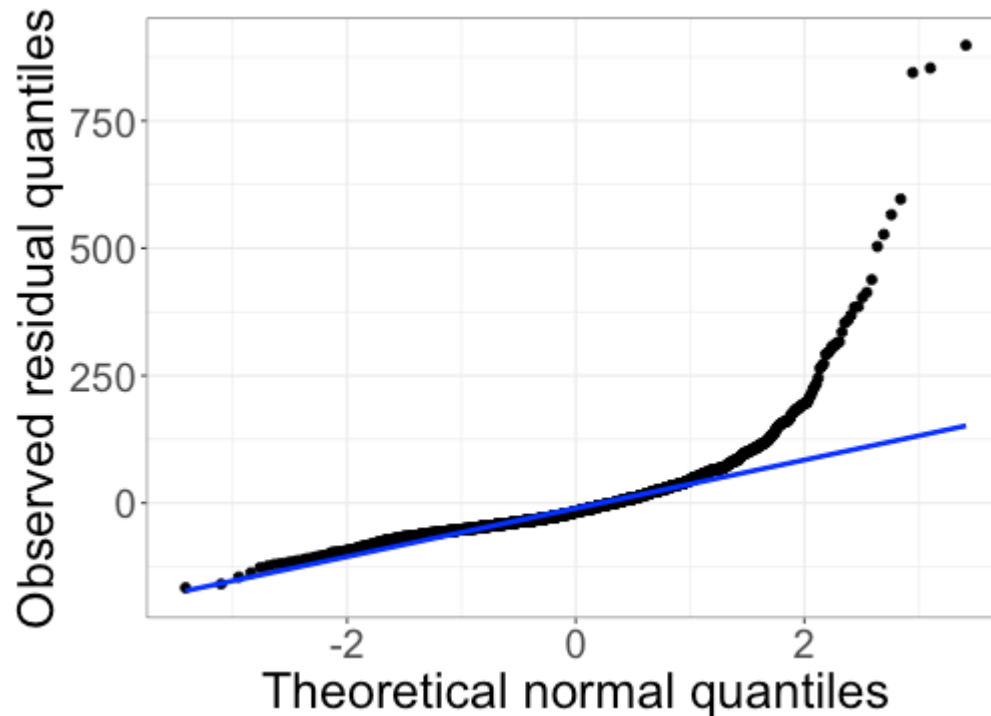
- + Check whether the residuals appear normal with a QQ plot

QQ plot for the residuals



Do the residuals appear normal?

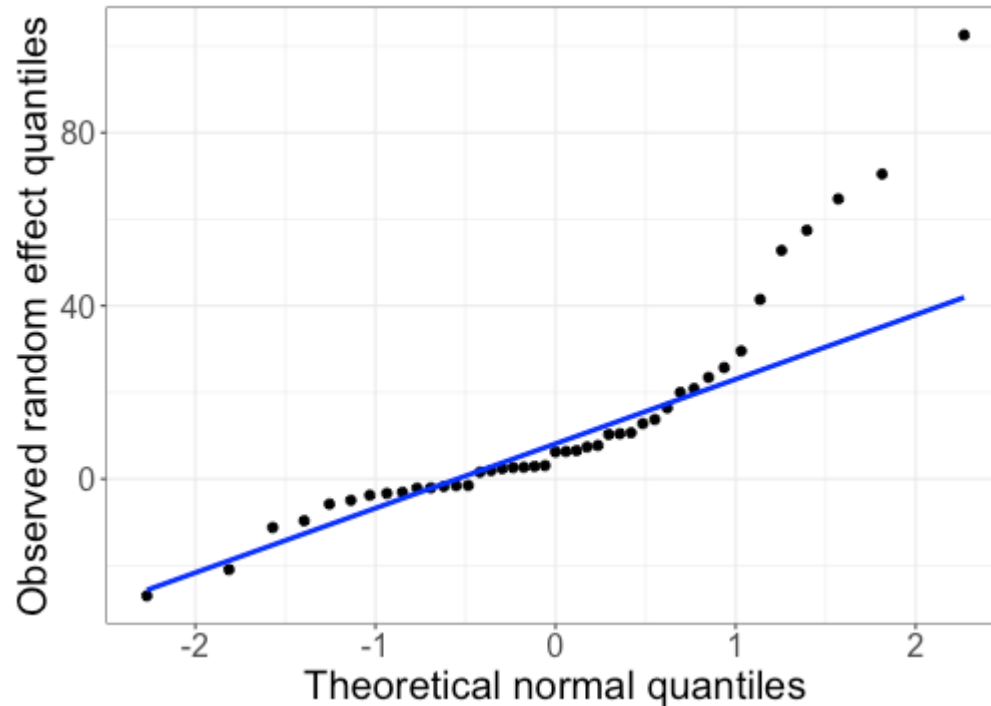
QQ plot for the residuals



Do the residuals appear normal?

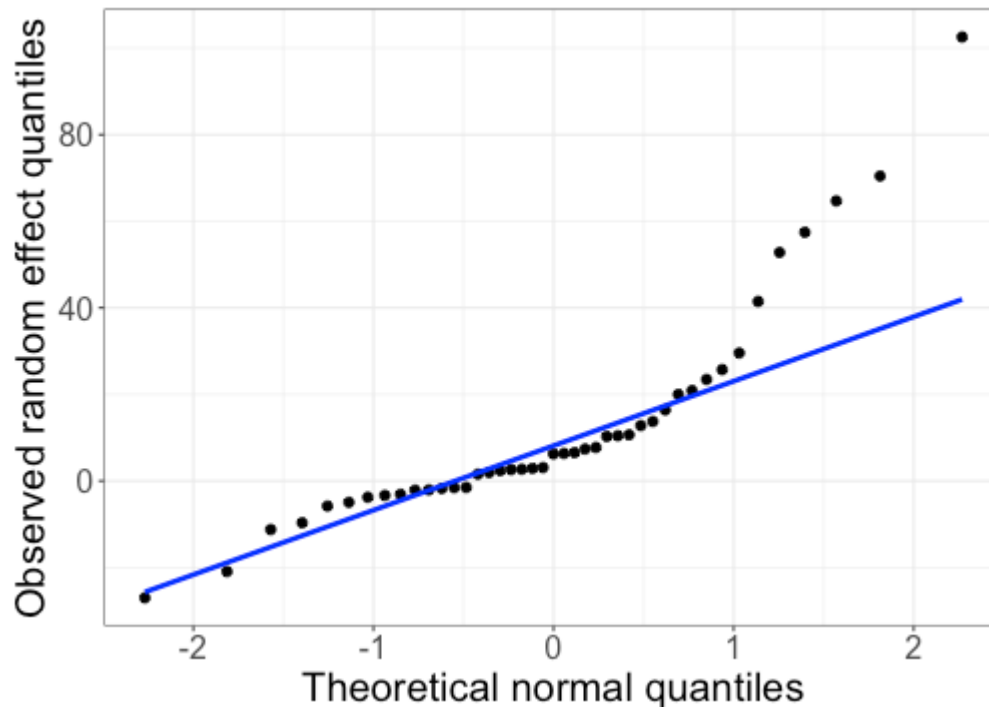
No -- though the sample size is large, so it might not matter

QQ plot for the random effects



Do the random effects appear normal?

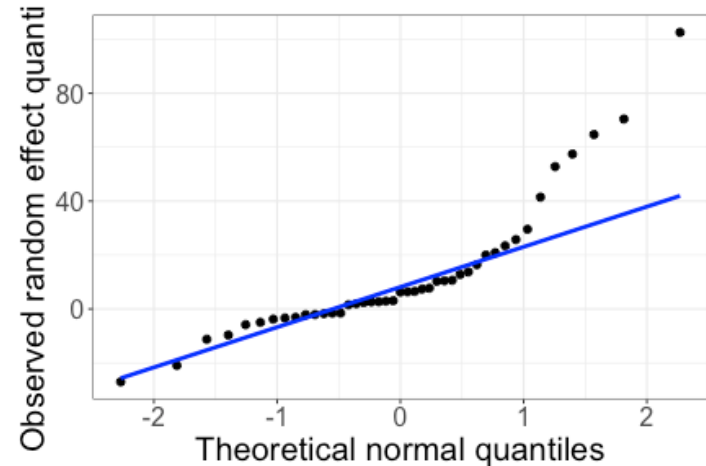
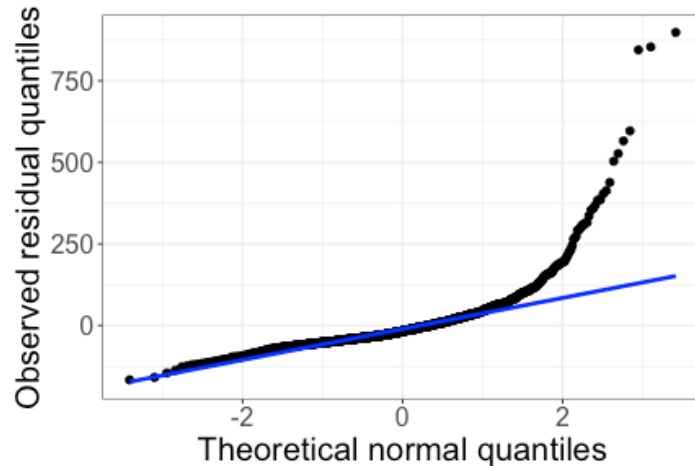
QQ plot for the random effects



Do the random effects appear normal?

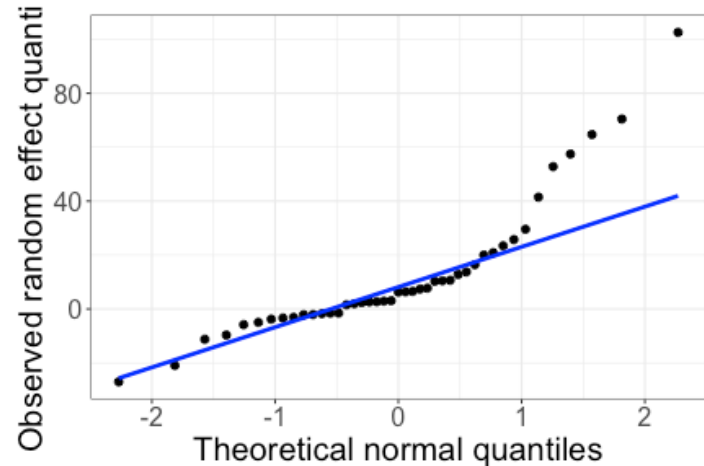
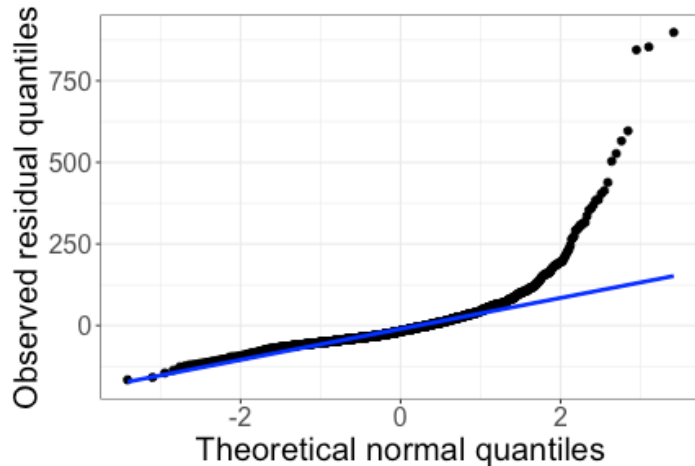
No -- and this could be more problematic

Addressing assumption violations



How could we address violations of the normality assumptions?

Addressing assumption violations



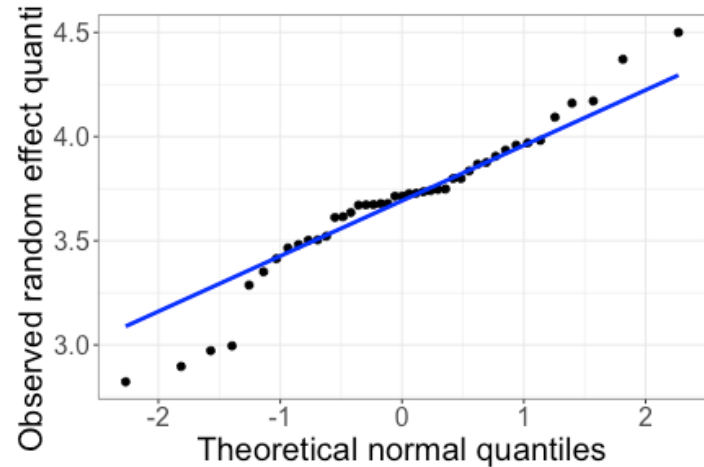
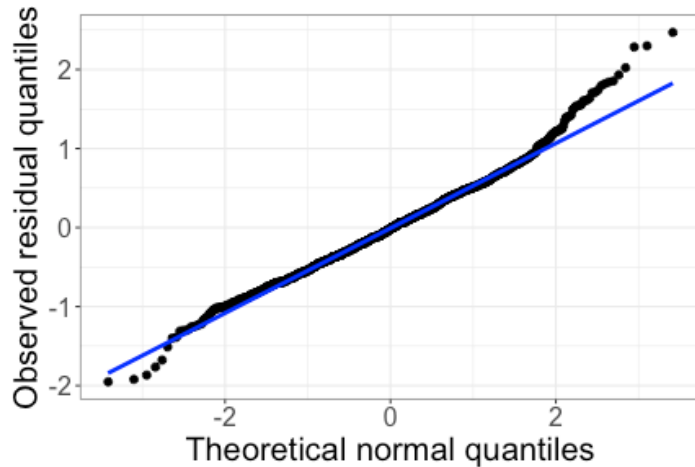
How could we address violations of the normality assumptions?

Transformations!

Transformations

$$\log(\text{Price}_{ij}) = \beta_0 + \beta_1 \text{Satisfaction}_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \stackrel{iid}{\sim} N(0, \sigma_u^2) \quad \varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2)$$



- ✚ Transformations can help with both the residuals and random effects

Class activity

Work on the class activity on the course website. Submit your HTML on Canvas when finished.